0–1.5 min / 1 A code can protect information and implement a gate Lead with the result.

1.5–3 min / 2 Binary codes describe a quantum encoding Assume binary linear codes, introduce only the required quantum notation.

3–5 min / 3 The eight-qubit cube starts with affine functions The picture labels vertices by u and shows the parity truth table, not a noisy readout.

5–7 min / 4 Complementary words encode three logical qubits Explain relative phase on the pair: changing the sign of only one word leaves the encoding subspace.

7–8.5 min / 5 Single-qubit phases can produce a three-qubit CCZ T is a phase rotation on physical basis bit 1.

8.5–10 min / 6 The toy gate works, but its distance is two For an affine function other than p and p+1, the Hamming distance to p is four.

10–11.5 min / 7 Overlap counts control weights of sums Use the face/edge intuition from the cube, but avoid a second running example.

11.5–13.5 min / 8 The general gate condition is a signed-weight congruence This is the quantum-to-classical bridge.

13.5–15.5 min / 9 Only single, pair, and triple overlaps are needed Derive the coefficients from the previous slide rather than reading a condition list.

15.5–17 min / 10 Geometry fixes the stabilizers before the search Separate design assumptions from necessary conditions.

17–19 min / 11 Projectivity rules out weight-one and weight-two Z logicals Give the short proof.

19–21 min / 12 The support is projective space minus a subspace Recall a point in binary projective space is simply a nonzero vector.

21–23 min / 13 Hyperplane sections give the two nonzero weights Show the two cases, which prove both rank six and eight-divisibility.

23–25 min / 14 SAT assigns three logical bits to each geometric point The count is parity constraints, not the number of ordinary CNF clauses; solver auxiliary variables depend on encoding.

25–26.5 min / 15 The published pattern is a layer of 48 phase gates Show the actual paper figure, not an independently invented pattern.

26.5–28 min / 16 A three-point witness makes the Z-distance exact No X check sees the three sites because the columns sum to zero.

28–30.5 min / 17 The lower bound turns supports into Reed–Muller words Give the mechanism, not the full classification.

30.5–33 min / 18 Our exclusion argument covers every length through 38 Describe analytic reductions plus finite enumeration honestly.

33–35 min / 19 Concurrent classification closes the 39–46 gap The original preprint left 39 through 46 open; the seminar reflects the concurrent classification.

35–37 min / 20 The geometric construction produces a family This is a family of finite constructions, not an asymptotic threshold claim.

37–38.5 min / 21 Gate-aware search complements general SAT code design Keep the comparison factual, not a novelty competition.

38.5–40 min / 22 Open questions connect algebra to implementation Return to the promised bridge between classical codes and logical quantum gates.

40 minutes: finish and invite questions.